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Question Paper Code : 14963

B.E / B.TECH. DEGREE EXAMINATION, NOVEMBER / DECEMBER 2018

Third Semester

Common to B.E. – Computer Science and Engineering & B.Tech. – Information Technology

13CS303 – COMPUTER ORGANIZATION AND ARCHITECTURE

(Regulations: Mepco – R2013)

Duration: 3 Hours Max. : 100 Marks

Answer ALL Questions

PART A – (10 × 2 = 20 Marks)

1. _____ style processors are characterized by "Simple addressing modes, All instructions fitting in a single word".

   A) CISC B) RISC C) Vector D) All the above

2. Number of logic gates needed to build a 4-bit carry-lookahead adder is

   A) 31 B) 30 C) 29 D) 28

3. In the datapath, for selecting a particular data – either content of register or the immediate value, _____ hardware is used.

   A) Demultiplexer B) Multiplexer C) Encoder D) Decoder

4. Compare hardwired control and microprogrammed control.

5. The kind of dependency that exist among the following instructions is _____.

   Add R2, R3, #100

   Subtract R9, R2, #30

   A) Control dependency B) Name dependency

   C) Data dependency D) None of the above

6. Give the basic performance equation.

7. The memory that involves only reading of the stored data is

   A) RAM B) Secondary storage

   C) Virtual memory D) ROM

8. Define cache hit and cache miss.

9. Compare synchronous and asynchronous bus.

10. State the significance of PCI bus.

PART B – (5 × 16 = 80 Marks)

11. a) i. Perform the following multiplication operation using Booth algorithm and explain: 13 * (-6). (8 Marks)

11. a) ii. List and explain various addressing modes supported by RISC-type processors. (8 Marks)

OR

11. b) i. State the rules for floating-point addition and multiplication operations and do the following as mentioned:

A) 0.00123 + 34.892

B) 45.821 * 0.000648 (8 Marks)

11. b) ii. Give brief account on straight-line sequencing and branching. (8 Marks)

12. a) i. Find the actions involved in fetching and executing the following RISC-style instruction:

Load R3, X(R6).

Specify the steps involved in completing the Load instruction. (8 Marks)

12. a) ii. What are the two approaches for generating the control signals? Explain in detail about hardwired control. (8 Marks)

OR

12. b) i. Specify the sequence of actions needed to fetch and execute the following CISC style instruction and explain.

And X(R1), R2 (8 Marks)

12. b) ii. Describe about the datapath in a RISC style processor with a neat diagram. (8 Marks)

13. a) i. Consider the following instructions at the given addresses in the memory:

1000 Add R3, R2, #20

1004 Subtract R5, R4, #3

1008 And R6, R4, #0x3A

1012 Add R7, R2, R4

Initially, registers R2 and R4 contain 2000 and 50, respectively. These instructions are executed in a computer that has a five-stage pipeline. The first instruction is fetched in clock cycle 1, and the remaining instructions are fetched in successive cycles.

A) Draw a diagram that represents the flow of the instructions through the pipeline. Describe the operation being performed by each pipeline stage during clock cycles 1 through 8.

B) Describe the contents of registers IR, PC, RA, RB, RY, and RZ in the pipeline during cycles 2 to 8. (8 Marks)

13. a) ii. Why do we need branch prediction? Explain various methods of dynamic branch prediction with neat diagram. (8 Marks)

OR

13. b) i. The forwarding path allows the contents of register RZ to be used directly in an ALU operation. The result of that operation is stored in register RZ, replacing its previous contents. This problem involves tracing the contents of register RZ over multiple cycles. Consider the two instructions,

I1: Add R3, R2, R1

I2: LShiftL R3, R3, #1

While instruction I1 is being fetched in cycle 1, a previously fetched instruction is performing an ALU operation that gives a result of 17. Then, while instruction I1 is being decoded in cycle 2, another previously fetched instruction is performing an ALU operation that gives a result of 198. Also, during cycle 2, registers R1, R2, and R3 contain the values 30, 100, and 45, respectively. Using this information, draw a timing diagram that shows the contents of register RZ during cycles 2 to 5. (8 Marks)

13. b) ii. How the pipeline is organized with five stages in a RISC style processor? Describe it with appropriate diagram. (8 Marks)

14. a) i. How can you construct larger memories by using smaller memory modules? Construct an 8M x 32 memory using 512K x 8 memory chips. (8 Marks)

14. a) ii. How the virtual address is translated into physical address? Explain with necessary diagram. (8 Marks)

OR

14. b) i. A block-set-associative cache consists of a total of 64 blocks, divided into 4-block sets. The main memory contains 4096 blocks, each consisting of 32 words. Assuming a 32-bit byte-addressable address space, how many bits are there in each of the Tag, Set, and Word fields? (4 Marks)

14. b) ii. To construct a 16M x 32 memory, how many 1M x 4 memory chips are needed? Explain. (4 Marks)

14. b) iii. Explain in detail about mapping functions of cache memory. In each kind of mapping method, explain how the main memory address is divided into various components? (8 Marks)

15. a) Explain the following in detail:

A) Interrupts

B) Program controlled I/O (16 Marks)

OR

15. b) Explain the following in detail:

A) Arbitration

B) Universal Serial Bus (USB) – an interconnection standard (16 Marks)